Oxygen Cylinder



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Oxygen Cylinder; Oxygen Cylinder with Connector, Key, Face Mask and Tubing

What it is used for

An oxygen cylinder supplies medical oxygen through a compatible regulator, flow control, tubing, and delivery interface when prescribed or authorised. Correct cylinder handling, leak checks, flow settings, and fire precautions are essential because oxygen supports rapid combustion and a high-pressure cylinder can cause serious injury if mishandled.

Main parts

- Oxygen cylinder stores oxygen under high pressure.
- Cylinder valve controls the gas outlet.
- Regulator reduces cylinder pressure to a usable delivery pressure.
- Flowmeter or flow control sets the prescribed oxygen flow.
- Outlet connector links the regulator to the delivery tubing.
- Oxygen tubing carries oxygen to the delivery interface.
- Face mask delivers oxygen to the patient where prescribed.
- Cylinder key or wrench operates the valve where the design requires one.
- Cylinder trolley or restraint secures the cylinder during movement and use.

Before you start

1. Only trained and authorised personnel should set up or administer medical oxygen.
2. Confirm the patient's identity, prescribed oxygen delivery method, and flow rate with the authorised clinician.
3. Check the cylinder label, oxygen identification, pressure status, service condition, and available contents.
4. Inspect the cylinder, valve, regulator, flowmeter, connectors, tubing, and face mask for damage or contamination.
5. Confirm the regulator is designed for the cylinder and that all connectors are compatible.
6. Secure the cylinder upright using an approved stand or trolley before opening the valve.
7. Keep the area free from flames, smoking, oil, grease, and other ignition or combustible hazards.

How to use

1. Wash hands and prepare the prescribed oxygen delivery equipment.
2. Secure the cylinder and confirm the regulator and flow control are fitted correctly.
3. Ensure the flow control is in the specified initial position before opening the cylinder valve.
4. Open the cylinder valve slowly according to the regulator instructions and check for leaks using an approved method.

5. Set the prescribed flow rate and confirm oxygen is flowing through the delivery tubing.
6. Connect the tubing to the prescribed face mask or other authorised delivery interface.
7. Fit the interface to the patient as directed by the clinician and check comfort and correct positioning.
8. Monitor the patient's breathing, oxygen saturation where indicated, and response to therapy according to the care plan.
9. If the supply fails, a leak occurs, or the patient's condition worsens, follow the clinical emergency procedure immediately.
10. When therapy ends, close the cylinder valve according to the instructions and release downstream pressure only as directed by the manufacturer.
11. Disconnect and clean or dispose of patient-contact items according to infection-control procedures.
12. Record the flow rate, start and stop times, patient response, cylinder status, and any faults.

⚠ Safety

- Never use oxygen near smoking, open flames, sparks, or heat sources.
- Never apply oil, grease, petroleum products, or unapproved lubricants to oxygen equipment.
- Keep the cylinder secured upright and protect the valve from impact or damage.
- Do not use a leaking, damaged, unidentified, or out-of-service cylinder or regulator.
- Do not change the prescribed oxygen flow rate without clinical authorisation.
- Do not attempt to repair the cylinder valve or regulator; follow the emergency procedure for leaks or equipment failure.

Care & storage

- Store cylinders in a designated, ventilated area away from heat and combustible materials.
- Keep full and empty cylinders identified and stored according to facility procedures.
- Inspect the cylinder, regulator, flowmeter, tubing, and connectors before each use.
- Clean reusable equipment using approved methods and follow infection-control requirements for patient-contact components.
- Arrange authorised servicing and return cylinders through the approved supplier or facility process.

Common mistakes to avoid

- Using an incompatible regulator or connector on the oxygen cylinder.
- Leaving the cylinder unsecured or opening its valve too quickly.
- Changing oxygen flow without clinical authorisation.
- Allowing oil, grease, flames, or smoking near oxygen equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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